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Modeling & Simulation Lifecycle



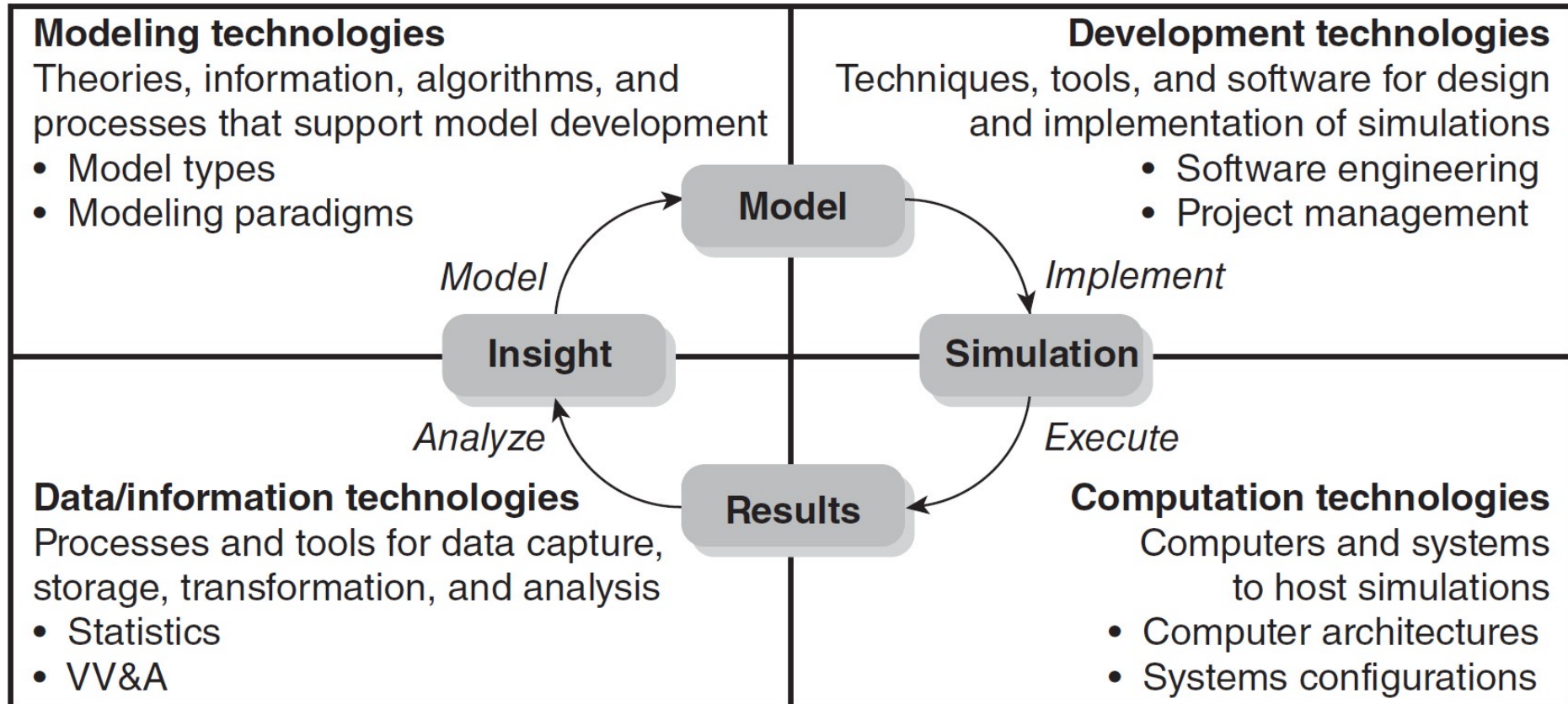
M&S Lifecycle

- M&S as a process
- Basic M&S lifecycle
 - Modelling
 - Development
 - Execution
 - Analysis
- Detailed M&S lifecycle steps

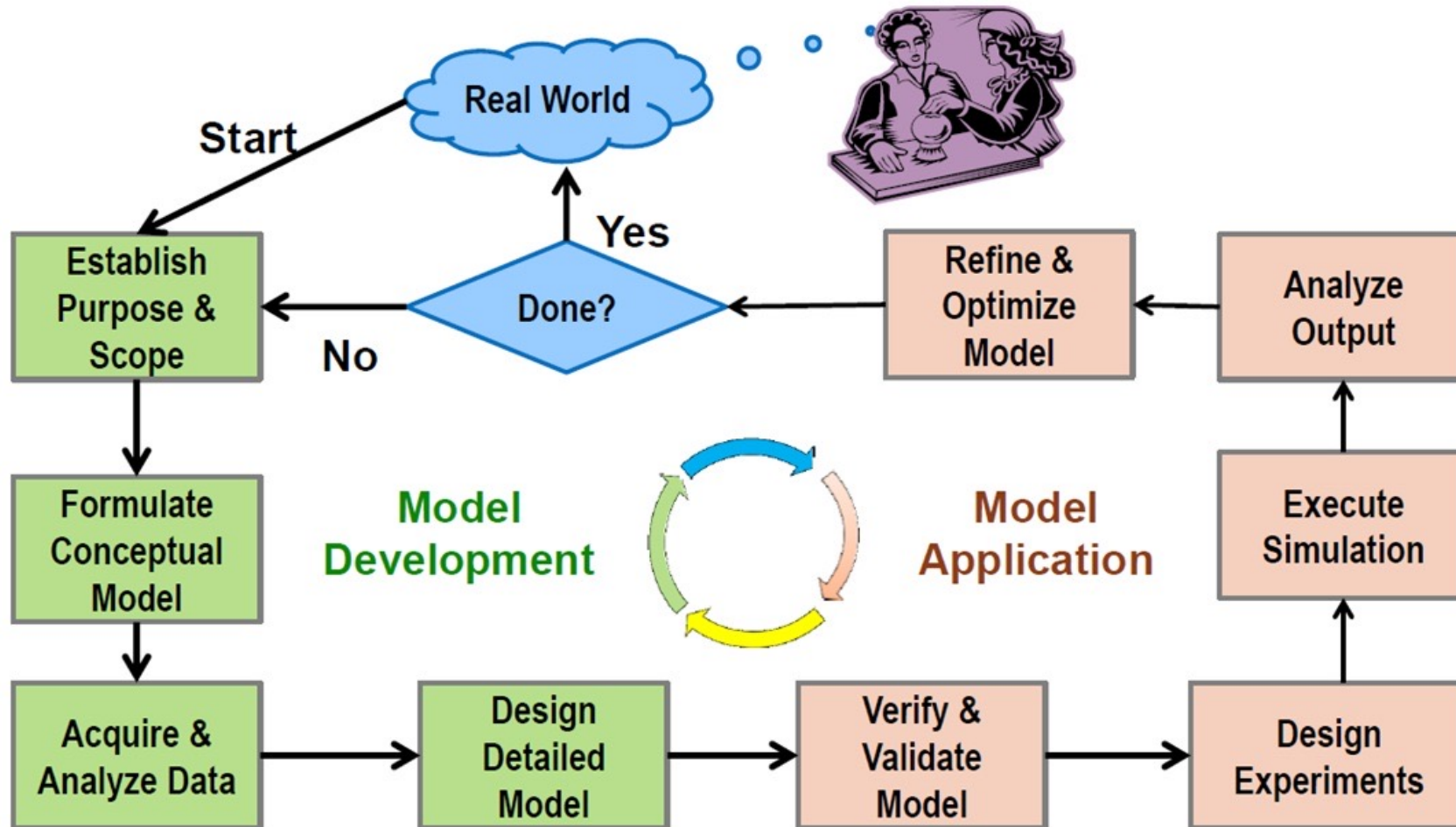
M&S as a process

- M&S definition – repetition
- Formal/Informal M&S process
- Shaped based on the purpose
 - Aid for thought
 - Aid to communicate
 - Aid for education and training
 - Aid for experimentation
 - Aid for decision
 - Aid for prediction

M&S as a Process

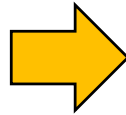


M&S as a Process



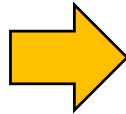
ESTABLISH PURPOSE AND SCOPE

- Problem formulation



- Program Goals
- Identify User/Sponsor Needs

- System definition



- Existing Domain Descriptions
- Info on available Resources

- Limitation of Simulation
- Budget
- Team
- Management

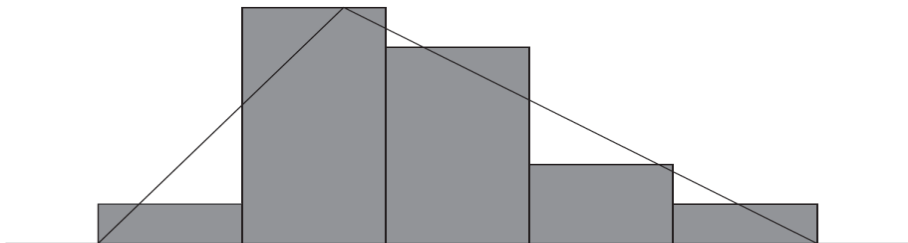


Develop Conceptual Model

- To help stakeholders understand what is going to be modeled
- Transform human ideas into the uniform representation/description
- Abstraction from the real system – creates a frame of reference
- Description is done in informal or formal way
- Abstraction - transformation of essential features of real system into a different form

Acquire and Analyze and model data

- Conceptual model gives us the answer about structure, entities, boundaries, processes in the form of mathematical or formal representation, but we don't have any information about data
- Addressed System/Process/Phenomenon (**Simuland**) exists and is accessible -> Gather samples
 - Use directly sample as input
 - Generate empirical distribution
 - Use theoretical distribution
- Simuland doesn't exist -> Use subject matter experts (**SMEs**)
 - Use theoretical distribution (uniform and triangle)



Created histograms are used to identify the family of distribution it belongs to.. And then estimate parameters of distribution – sample mean and sample variance **Goodness of fit test (Chi-Square Test)**